

Waves and Wave Scattering

Notes taken by: Kaichao Xu

Bases on the lectures by Prof V Smyshlyaev

January 2025

Contents

1	Acoustic Waves	2
1.1	Irrotational fields and velocity potential	2
1.2	Spherically symmetric waves	3
1.3	Time-Harmonic Waves	3
1.4	Causality and Sommerfeld Radiation Condition	5
1.5	Acoustic energy, Intensity	6
2	Electromagnetic Waves	7
3	Boundary and Interface Conditions	9
3.1	Acoustic BCs	9
3.2	EM Interface/Boundary Conditions	9
4	Canonical Cases	10
4.1	Reflection by a plane (acoustics)	10
4.2	Reflection of EM waves by P.C. Plane	10
4.3	Reflection and Refraction by a plane EM Interface	11
4.4	Total Internal Reflection (TIR)	12
4.5	Rapidly Pulsating Spherical Sources	13
5	Green's functions, Green's formulas	14
5.1	Points Sources	14
5.2	Green's Functions	15
5.3	Kirchhoff-Helmholz Integral Theorem	15

1 Acoustic Waves

Some important equations:

$$\begin{aligned}\frac{1}{c^2} \frac{\partial p}{\partial t} + \rho_0 \nabla \cdot \underline{u} &= 0 \\ \frac{\partial \underline{u}}{\partial t} &= -\frac{1}{\rho_0} \nabla p \\ \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} - \nabla^2 p &= 0\end{aligned}$$

Plane waves: $p(\underline{x}, t) = p(x, y, z, t) = p(x, t)$ thus,

$$\frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} - \frac{\partial^2 p}{\partial x^2} = 0$$

General solution of the above equation is given by

$$p(x, t) = f(x - ct) + g(x + ct)$$

with $f, g \in C^2$. More generally, for all "direction" $\underline{n}$ with $|\underline{n}| = 1$,

$$p(\underline{x}, t) = f(\underline{x} \cdot \underline{n} - ct) + g(\underline{x} \cdot \underline{n} + ct)$$

for all $f, g \in C^2$ solves $\frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} - \nabla^2 p = 0$. In this equation, $p_1(\underline{x}, t) = f(\underline{x} \cdot \underline{n} - ct)$ describes a plane wave in positive $\underline{n}$ -direction i.e. any plane P orthogonal to $\underline{n}$ and p_1 is constant on P so $\underline{x} \cdot \underline{n} - ct = A$ which plane of the constant value A moves in direction $\underline{n}$ with wave speed c ("phase velocity" c). Similarly, $p_2 = g(\underline{x} \cdot \underline{n} + ct)$ describes plane wave in negative $\underline{n}$ direction.

Example. Consider $p(\underline{x}, t) = f(3x - 4z + 5ct)$ then the direction is

$$\underline{n} = \frac{(3, 0, -4)}{\|(3, 0, -4)\|} = \left(\frac{3}{5}, 0, -\frac{4}{5}\right)$$

and since

$$p = f(5(\underline{n} \cdot \underline{x} + ct)) = F(\underline{x} \cdot \underline{n} + ct)$$

so phase velocity is given by c .

Example. $p(\underline{x}, t) = \cos(\underline{x} \cdot \underline{n} - ct)$.

1.1 Irrotational fields and velocity potential

Let at $t = 0$, the velocity $\underline{u}$ is irrotational i.e. $\nabla \times \underline{u} = 0$ at $t = 0$. Taking curl of

$$\frac{\partial \underline{u}}{\partial t} = -\frac{1}{\rho_0} \nabla p$$

then

$$\nabla \times \frac{\partial \underline{u}}{\partial t} = \frac{\partial}{\partial t} (\nabla \times \underline{u}) = \nabla \times \left(-\frac{1}{\rho_0} \nabla p\right) = -\frac{1}{\rho_0} \nabla \times (\nabla p) = 0$$

Therefore, $\frac{\partial}{\partial t} (\nabla \times \underline{u}) = 0$ so $\nabla \times \underline{u} = 0$ for all t . There exists some $\phi(\underline{x}, t)$ (velocity potential) such that $\underline{u} = \nabla \phi$.

Remark. ϕ is determined up to an arbitrary function $C(t)$ i.e. if $\psi = \phi + C(t)$ then ψ is another solution.

Then we have

$$\begin{aligned}\frac{\partial}{\partial t} (\nabla \phi) &= -\frac{1}{\rho_0} \nabla p \\ \implies \nabla \left(\frac{\partial \phi}{\partial t} + \frac{1}{\rho_0} p \right) &= 0\end{aligned}$$

so $\frac{\partial \phi}{\partial t} + \frac{1}{\rho_0} p = A(t)$ for some $A(t)$.

Remark. WLOG, $A(t) = 0$. Since

Note that

$$\frac{\partial \phi}{\partial t} - A(t) + \frac{1}{\rho_0} p = 0$$

$$\phi'(t) = \phi(t) - \underbrace{\int_0^t A(t') dt'}_{C(t)}$$

$$p = -\rho_0 \frac{\partial \phi}{\partial t} \Leftrightarrow \frac{\partial \phi}{\partial t} = -\frac{1}{\rho_0} p$$

differentiating this in t gives

$$\frac{\partial^2 \phi}{\partial t^2} = -\frac{1}{\rho_0} \frac{\partial p}{\partial t}$$

Since we know that $\frac{1}{c^2} \frac{\partial p}{\partial t} + \rho_0 \nabla \cdot \underline{u} = 0$, then we must have

$$\frac{\partial^2 \phi}{\partial t^2} = -\frac{1}{\rho_0} (-c^2 \rho_0 \nabla \cdot \underline{u}) = c^2 \nabla \cdot \nabla \phi = c^2 \nabla^2 \phi$$

Thus,

$$\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} - \nabla^2 \phi = 0 \quad (1)$$

i.e. velocity potential ϕ solves the same wave equation as p . If ϕ solves the wave equation then $p := -\rho_0 \frac{\partial \phi}{\partial t}$ solves the original two equations.

Exercise. Derive equation similar to ϕ when $\underline{u}$ is not irrotational.

1.2 Spherically symmetric waves

Seek solutions of wave equations

$$\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} - \nabla^2 \phi = 0$$

and p which are spherically symmetric, i.e.

$$p(\underline{x}, t) = p(|\underline{x}|, t) = p(r, t), r := |\underline{x}|$$

so constant on any sphere centred at 0. In spherical coordinates (r, θ, φ) , the Laplace operator is given by

$$\nabla^2 p = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial p}{\partial r}) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial p}{\partial \theta}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 p}{\partial \varphi^2}$$

Exercise. If $p = p(r)$ and $r = |\underline{x}|$ then we have

$$\nabla^2 p = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial p}{\partial r})$$

For $p(\underline{x}, t) = p(r, \theta, \varphi) = p(r, t)$ then

$$\frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} - \frac{1}{c^2} \frac{\partial}{\partial r} (r^2 \frac{\partial p}{\partial r}) = 0$$

Try $p = \frac{q}{r}$ then

$$\begin{aligned} \frac{\partial p}{\partial r} &= -\frac{1}{r^2} q + \frac{1}{r} \frac{\partial q}{\partial r} \\ r^2 \frac{\partial p}{\partial r} &= -q + r \frac{\partial q}{\partial r} \\ \frac{\partial}{\partial r} (r^2 \frac{\partial p}{\partial r}) &= -\frac{\partial q}{\partial r} + \frac{\partial q}{\partial r} + r \frac{\partial^2 q}{\partial r^2} \end{aligned}$$

Therefore, we must have

$$\begin{aligned} \frac{1}{c^2} \frac{1}{r} \frac{\partial^2 q}{\partial t^2} - \frac{1}{c^2} \frac{\partial^2 q}{\partial r^2} &= 0 \\ \implies \frac{1}{c^2} \frac{\partial^2 q}{\partial t^2} - \frac{\partial^2 q}{\partial r^2} &= 0 \end{aligned}$$

i.e. g solves the 1-D wave equation, with $p \rightarrow q$ and $x \rightarrow r$. The general solution is given by

$$\begin{aligned} q(r, t) &= f(r - ct) + g(r + ct), \quad \forall f, g \in C^2 \\ \implies p &= \frac{f(r - ct)}{r} + \frac{g(r + ct)}{r} \end{aligned}$$

Remark.

- (1) f -part: $p_1 = \frac{f(r-ct)}{r}$, p_1 is constant on any sphere centred at 0. e.g. for "nodal points" $p_1 = 0 \Leftrightarrow r - ct = A$ such that $f(A) = 0$. The sphere inflates then p_1 is an outgoing wave.
- (2) For $p_2 = \frac{g(r+ct)}{r}$, the "nodal sphere" deflates so incoming wave.

1.3 Time-Harmonic Waves

Seek solutions of wave equations

$$\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} - \nabla^2 \phi = 0$$

and p oscillating sinusoidally in time

$$p(\underline{x}, t) = A(\underline{x}) \cos(\omega t - \Psi(\underline{x}))$$

We denote $A(\underline{x})$ by amplitude and $\Psi(\underline{x})$ is known as the phase. $\omega > 0$ is called the angular frequency. $T = \frac{2\pi}{\omega}$ is known as time period and

$$p(\underline{x}, t) = \text{Re}\{e^{-i\omega t + i\Psi(\underline{x})}\}$$

Denote $A(\underline{x})e^{i\Psi(\underline{x})}$ by $\hat{p}(\underline{x})$. Then we have

$$p(\underline{x}, t) = \text{Re}\{\hat{p}(\underline{x})e^{-i\omega t}\}$$

and $\hat{p}(\underline{x})$ is called "complex amplitude". Note that $|\hat{p}| = A$ and $\arg \hat{p} = \Psi$ we immediately have

$$\begin{aligned} \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \text{Re}\{\hat{p}(\underline{x})e^{-i\omega t}\} - \nabla^2 \text{Re}\{\hat{p}(\underline{x})e^{-i\omega t}\} &= 0 \\ \text{Re}\{\frac{1}{c^2} (-i\omega)^2 e^{-i\omega t} \hat{p} - (\nabla^2 \hat{p}) e^{-i\omega t}\} &= 0 \end{aligned}$$

$$\underbrace{\operatorname{Re}\{(\nabla^2 \hat{p} + \frac{\omega^2}{c^2} \hat{p})e^{-i\omega t}\}}_S = 0 \text{ for all } t$$

Then

$$(\operatorname{Re} S) \cos \omega t + (\operatorname{Im} S) \sin \omega t = 0 \text{ for all } t$$

we immediately have $S = 0$ by evaluating the value of t at 0 and $\frac{\pi}{2\omega}$. This indicates

$$\nabla^2 \hat{p} + k^2 \hat{p} = 0$$

where $k := \frac{\omega}{c}$ is the wavenumber. This equation is called Helmholtz equation and is the main equation for Time-Harmonic waves.

Velocity is also sought in Time-Harmonic form

$$\underline{u}(\underline{x}, t) = \operatorname{Re}\{\hat{\underline{u}}(\underline{x})e^{-i\omega t}\}$$

Then we must have

$$\begin{aligned} \operatorname{Re}\{(-i\omega)\hat{\underline{u}}e^{-i\omega t}\} &= -\frac{1}{\rho_0} \operatorname{Re}\{(\nabla \hat{p})e^{-i\omega t}\} \\ \implies \operatorname{Re}\{(-i\omega \underline{u} + \frac{1}{\rho_0} \nabla \hat{p})e^{-i\omega t}\} &= 0 \text{ for all } t \\ \implies \hat{\underline{u}} &= -\frac{i}{\rho_0 \omega} \nabla \hat{p} \end{aligned}$$

Exercise. If $\hat{p}$ solves Helmholtz equation, and $\hat{\underline{u}} = -\frac{i}{\rho_0 \omega} \nabla \hat{p}$ then $p := \operatorname{Re}\{\hat{p}e^{-i\omega t}\}$ and $\underline{u} = \operatorname{Re}\{\hat{\underline{u}}e^{-i\omega t}\}$ solves the original two equations.

Remark. T-H fields are automatically irrotational.

$$\begin{aligned} \underline{u} &= \operatorname{Re}\{\hat{\underline{u}}e^{-i\omega t}\} = \operatorname{Re}\{-\frac{i}{\rho_0 \omega} \nabla \hat{p}e^{-i\omega t}\} \\ &= \nabla(\operatorname{Re}\{-\frac{i}{\rho_0 \omega} \hat{p}e^{-i\omega t}\}) = \nabla \phi \end{aligned}$$

where $\phi = \operatorname{Re}\{\hat{\phi}e^{-i\omega t}\}$, $\hat{\phi} = -\frac{i}{\rho_0 \omega} \hat{p}$.

Plane T-H waves: for a plane T-H wave in positive $\underline{n}$ -direction with $|\underline{n}| = 1$. Then

$$p = f(\underline{x} \cdot \underline{n} - ct) = \operatorname{Re}\{\hat{p}(\underline{x})e^{-i\omega t}\} = \operatorname{Re}\{e^{i\frac{\omega}{c}(-ct + \underline{x} \cdot \underline{n})} A\}$$

where $A := \hat{p}(\underline{x})e^{-ik\underline{n} \cdot \underline{x}} = F(\underline{x} \cdot \underline{n} - ct)$ is a constant. Thus, $p = |A| \cos(\omega t - \underline{k} \cdot \underline{x} - \psi) = \operatorname{Re}\{\hat{p}e^{-i\omega t}\}$ where $\hat{p} = Ae^{ik \cdot \underline{x}}$ and $\underline{k} = k\underline{n}$ is wavevector. So this is the general form of plane T-H waves. $\lambda = \frac{2\pi}{k}$ is wavelength. Recall $T = \frac{2\pi}{\omega}$ is the time period, $f := \frac{1}{T} = \frac{\omega}{2\pi}$ is the frequency. And $\lambda f = c$.

Now for plane T-H velocity fields:

$$\begin{aligned} \hat{\underline{u}} &= -\frac{i}{\rho_0 \omega} \nabla \hat{p} = -\frac{i}{\rho_0 \omega} \nabla (Ae^{ik \cdot \underline{x}}) = -\frac{iA}{\rho_0 \omega} i \underline{k} e^{ik \cdot \underline{x}} = \frac{Ak\underline{n}}{\rho_0 \omega} e^{-k \cdot \underline{x}} = \frac{\underline{n}}{\rho_0 c} Ae^{ik \cdot \underline{x}} = \frac{\underline{n}}{\rho_0 c} \hat{p} \\ \implies \hat{\underline{u}} &= \operatorname{Re}\{\hat{\underline{u}}e^{-i\omega t}\} = \frac{\underline{n}}{\rho_0 c} \operatorname{Re}\{\hat{p}e^{-i\omega t}\} \end{aligned}$$

i.e. $\hat{\underline{u}} = \frac{\underline{n}}{\rho_0 c} \hat{p} = \frac{\underline{n}}{\rho_0 c} |A| \cos(\omega t - \underline{k} \cdot \underline{x} - \psi)$. So $\underline{u}$ is "in phase" with p coinciding in maxima and minima and $\underline{u}$ is along $\underline{n}$, direction of wave propagation/phase velocity. Acoustic-wave are Longitudinal (not transverse).

Remark. Assuming sufficient decay as $t \rightarrow \pm\infty$, any acoustic field can be expressed as a superposition of T-H waves via Fourier Transform in time: If $p(\underline{x}, t)$ solves the equation, it can be represented via inverse Fourier Transform of $p^*(\underline{x}, \omega)$:

$$p(\underline{x}, t) = (2\pi)^{-1/2} \int_{-\infty}^{\infty} p^*(\underline{x}, \omega) e^{-i\omega t} d\omega$$

where (inversion formula):

$$p^*(\underline{x}, \omega) = (2\pi)^{-1/2} \int_{-\infty}^{\infty} p(\underline{x}, t) e^{i\omega t} dt$$

Now, $\frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} - \nabla^2 p = 0$ becomes

$$\begin{aligned} &= (2\pi)^{-1/2} \int_{-\infty}^{\infty} [\frac{1}{c^2} p^*(-i\omega)^2 e^{-i\omega t} - \nabla^2 p^*] e^{-i\omega t} d\omega \\ &= (-2\pi)^{-1/2} \int_{-\infty}^{\infty} (\nabla^2 p^* + \frac{\omega^2}{c^2} p^*) e^{-i\omega t} d\omega = 0 \end{aligned}$$

for all $t \in (-\infty, \infty)$. Now, by "inversion of zero-function", we see that

$$\nabla^2 p^* + k^2 p^* = 0$$

i.e. p^* solves Helmholtz equation.

$$\implies p(\underline{x}, t) = \int_{-\infty}^{\infty} \operatorname{Re}\{(2\pi)^{-1/2} p^*(\underline{x}, \omega) e^{-i\omega t}\} d\omega$$

this represents $p(\underline{x}, t)$ as a superposition of T-H waves $\text{Re}\{\hat{p}(\underline{x}, \omega)e^{-i\omega t}\}$, $\hat{p}(\underline{x}, \omega) := (2\pi)^{-1/2}p^*(\underline{x}, \omega)$. Note, if $\omega < 0$,
 $\text{Re}\{\hat{p}(\underline{x}, \omega)e^{-i\omega t}\} = \text{Re}\{\hat{p}(\underline{x}, \omega)e^{i\omega t}\}$

T-H wave with angular frequency $(-\omega) > 0$.

1.4 Causality and Sommerfeld Radiation Condition

Solutions of wave equations are often required to vanish before some time "source" switches on, no incoming waves from far way/infinity.

Causality: the effect (acoustic wave) can only follow the cause (source).

Remark. Wave equations are symmetric with respect to "future"- "past":

$$\frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} - \nabla^2 p = 0$$

stays the same if we replace $t \rightarrow -t$. So this equation has to be complemented by an additional "causality" condition. Consider a spherically symmetric wave:

$$p(r, t) = \frac{f(r - ct)}{r} + \frac{g(r + ct)}{r}, \quad f, g \in C^2$$

Outside the source region $r \leq r_0$, we then require: for any $r \geq r_0$ and for all $t \leq t_0$ then $p(r, t) \equiv 0$. For a general Spherical-symmetric solution,

$$p(\underline{x}, t) = \frac{f(r - ct)}{r}, \quad f \in C^2$$

Exercise. $g(\xi)$ is a constant, which can be "also allocated" to f .

For Spherical-symmetric T-H waves, we have

$$p(r, t) = \text{Re}\left\{\frac{e^{i(\frac{\omega}{c})(-ct+r)}}{r}A + \frac{e^{-i(\frac{\omega}{c})(ct+r)}}{r}B\right\}$$

with $A, B \in \mathbb{C}$ and

$$p(r, t) = \text{Re}\left\{\left(A\frac{e^{ikr}}{r} + B\frac{e^{-ikr}}{r}\right)e^{-i\omega t}\right\}$$

$$\implies p(r, t) = \text{Re}\{\hat{p}(r)e^{-i\omega t}\} \text{ with } \hat{p}(r) = A\frac{e^{ikr}}{r} + B\frac{e^{-ikr}}{r}$$

is general form of Spherical-symmetric T-H wave. For physical spherical-sphere T-H solutions, we expect not to give an incoming past.

Sommerfeld Radiation Conditions (SRC):

$$r\left(\frac{\partial \hat{p}}{\partial r} - ik\hat{p}\right) \rightarrow 0 \text{ as } r \rightarrow \infty$$

generally uniformly for non Spherical-Symmetric case. Check:

$$\hat{p} = A\frac{e^{ikr}}{r} + B\frac{e^{-ikr}}{r}$$

$$\frac{\partial \hat{p}}{\partial r} = \frac{ikA}{r}e^{ikr} - \frac{A}{r^2}e^{ikr} - \frac{ikB}{r}e^{-ikr} - \frac{B}{r^2}e^{-ikr}$$

$$r\frac{\partial \hat{p}}{\partial r} = ikAe^{ikr} - \frac{A}{r}e^{ikr} - ikBe^{-ikr} - \frac{B}{r}e^{-ikr}$$

$$ikr\hat{p} = ikAe^{ikr} + ikBe^{-ikr}$$

Then we must have

$$-2iBe^{-ikr} - \frac{A}{r}e^{ikr} - \frac{B}{r}e^{-ikr} \rightarrow 0$$

Now, as $|e^{\pm ikr}| = 1$, we must have $B = 0$. Thus,

$$\hat{p} = \frac{A}{r}e^{ikr}, \quad A \in \mathbb{C}$$

Remark. More generally, solutions to Helmholtz equation with SRC over FT of causal solutions. Mathematically, SRC ensures uniqueness and existence of the solutions of Helmholtz equation.

Now for Spherical-Symmetric T-H outgoing

$$\hat{p} = \frac{A}{r}e^{ikr} \implies \underline{\hat{u}} = -\frac{i}{\rho_0\omega}\nabla\hat{p} = -\frac{i}{\rho_0\omega}\nabla\left(\frac{A}{r}e^{ikr}\right) = -\frac{iA}{\rho_0\omega}\left(-\frac{1}{r^2} + \frac{ik}{r}\right)e^{ikr}\underline{\hat{r}}$$

$$= \frac{Ak}{\rho_0\omega r}\left(1 + \frac{i}{kr}\right)e^{ikr}\underline{\hat{r}} = \frac{A}{\rho_0 cr}\left(1 + \frac{i}{kr}\right)e^{ikr}\underline{\hat{r}}$$

and $\underline{u} = \text{Re}\{\underline{\hat{u}}e^{-i\omega t}\}$.

Interpretation: $\underline{u}$ is along $\underline{\hat{r}}$; $\frac{A}{\rho_0 cr}e^{ikr}\underline{\hat{r}}$ is "in phase" with $\hat{p}$ (as this was for phase T-H waves compared to $\hat{p}$). $\frac{A}{\rho_0 cr}\left(\frac{i}{kr}\right)e^{ikr}\underline{\hat{r}}$ is "out of phase" with p for $\pi/2$ since $i = e^{i\frac{\pi}{2}}$.

Remark. For $kr \gg 1$ i.e. $r \gg k^{-1} = \frac{\lambda}{2\pi}$, $r \gg \lambda$ so at "many wavelengths away", the second term is negligible. Thus, $\underline{u} \approx \frac{p}{\rho_0 c}$ as for plane waves.

1.5 Acoustic energy, Intensity

Return back to

$$\begin{aligned} \frac{1}{c^2} \frac{\partial p}{\partial t} + \rho_0 \nabla \cdot \underline{u} &= 0 \\ \frac{\partial \underline{u}}{\partial t} &= -\frac{1}{\rho_0} \nabla p \end{aligned}$$

Then we have

$$\begin{aligned} \frac{1}{\rho_0 c^2} \frac{\partial p}{\partial t} + \nabla \cdot \underline{u} &= 0 \\ \rho_0 \frac{\partial \underline{u}}{\partial t} + \nabla p &= 0 \end{aligned}$$

Consider

$$\begin{aligned} p \left(\frac{1}{\rho_0 c^2} \frac{\partial p}{\partial t} + \nabla \cdot \underline{u} \right) &= 0 \\ \underline{u} \cdot \left(\rho_0 \frac{\partial \underline{u}}{\partial t} + \nabla p \right) &= 0 \\ \frac{1}{\rho_0 c^2} p \frac{\partial p}{\partial t} + \rho_0 \underline{u} \cdot \frac{\partial \underline{u}}{\partial t} + p \nabla \cdot \underline{u} + \underline{u} \cdot \nabla p &= 0 \\ \frac{1}{\rho_0 c^2} \frac{\partial}{\partial t} \left(\frac{p^2}{2} \right) + \rho_0 \frac{\partial}{\partial t} \left(\frac{1}{2} \underline{u} \cdot \underline{u} \right) + \nabla \cdot (p \underline{u}) &= 0 \\ \frac{\partial}{\partial t} \left(\underbrace{\frac{1}{2} \rho_0 |\underline{u}|^2 + \frac{p^2}{2 \rho_0 c^2}}_w \right) + \nabla \cdot \left(\underbrace{p \underline{u}}_{\underline{I}} \right) &= 0 \end{aligned}$$

Then $\frac{\partial w}{\partial t} + \nabla \cdot \underline{I} = 0$ where $w := \frac{1}{2} \rho_0 |\underline{u}|^2 + \frac{p^2}{2 \rho_0 c^2}$ and $\underline{I} := p \underline{u}$. Note that $\frac{1}{2} \rho_0 |\underline{u}|^2$ is the kinetic energy density, and $\frac{p^2}{2 \rho_0 c^2}$ is the potential energy density (of compressed medium). Thus, w is a total (kinetic+potential) acoustic energy intensity. Integrate the above equation over volume V bounded by S with unit outer norm $\underline{n}$ then

$$\frac{d}{dt} \left(\int_V w(\underline{x}, t) dV \right) = - \int_V \nabla \cdot \underline{I} dV = - \int_S \underline{I} \cdot \underline{n} dS$$

$\underline{I}$ must measure energy in/out flow through S . Namely, $\underline{I}$, the acoustic intensity, show both direction of energy flow, and $|\underline{I}|$ is amount of energy transported through a unit area orthogonal to $\underline{I}$ in unit time.

Examples.

- (1) Plane T-H wave in direction $\underline{n}$. $\underline{I} = p \underline{u}$, recall $p = \text{Re}\{A e^{ik\underline{n} \cdot \underline{x} - i\omega t}\}$ and $\underline{u} = \frac{p}{\rho_0 c} \underline{n}$ thus

$$\underline{I} = p \underline{u} = \frac{p^2}{\rho_0 c} \underline{n}$$

so energy flows in the direction of wave propagation $\underline{n}$ (of phase velocity) as expected.

- (2) S-S, T-H outgoing waves: $\hat{p} = \frac{A}{r} e^{ikr}$ where $A \in \mathbb{C}$.

$$p = \text{Re}\{\hat{p} e^{-i\omega t}\} = \text{Re}\left\{ \frac{|A|}{r} e^{ikr + i\Psi - i\omega t} \right\}$$

$$p = \frac{|A|}{r} \cos(\omega t - kr - \Psi)$$

Next, $\hat{\underline{u}} = \hat{u}_r \tilde{\underline{r}}$ where $\hat{u}_r = \frac{A}{\rho_0 c r} e^{ikr} \left(1 + \frac{i}{kr}\right)$.

$$\implies \hat{\underline{u}} = \text{Re}\{\hat{\underline{u}} e^{-i\omega t}\} = u_r \tilde{\underline{r}}$$

where $u_r = \text{Re}\left\{ \frac{|A|}{\rho_0 c r} e^{ikr + i\Psi} \left(1 + \frac{i}{kr}\right) e^{-i\omega t} \right\}$

$$= \frac{|A|}{\rho_0 c r} \cos(\omega t - kr - \Psi) + \frac{|A|}{\rho_0 c r^2 k} \sin\left(\underbrace{\omega t - kr - \Psi}_{\varphi}\right)$$

$$\implies \underline{I} = I_r \tilde{\underline{r}}$$

where

$$I_r = \frac{|A|^2}{\rho_0 c r^2} \cos^2(\varphi) + \frac{|A|^2}{\rho_0 c r^3 k} \cos(\varphi) \sin(\varphi)$$

Now, for time-averaged $\underline{I}$, over period $T = \frac{2\pi}{\omega}$.

$$\langle \underline{I} \rangle := \frac{1}{T} \int_0^T \underline{I}(x, t) dt$$

Thus, $\langle \underline{I} \rangle = \langle I_r \rangle \tilde{r}$ where

$$\langle I_r \rangle = \frac{|A|^2}{\rho_0 c r^2} \langle \cos^2 \varphi \rangle + \frac{|A|^2}{\rho_0 k c r^3} \langle \sin \varphi \cos \varphi \rangle$$

Now,

$$\langle \cos^2 \varphi \rangle = \frac{1}{2} \int_0^T \cos^2(\omega t - kr - \Psi) dt = \frac{1}{2}$$

and $\langle \sin \varphi \cos \varphi \rangle = 0$. So

$$\langle \underline{I} \rangle = \frac{|A|^2}{2\rho_0 c r^2} \tilde{r}$$

so the time-averaged acoustic intensity is in outer radial direction as expected.

2 Electromagnetic Waves

The electromagnetic (EM) fields consist of two interacting vector fields. $\underline{E}(\underline{x}, t)$ electric field, $\underline{B}(\underline{x}, t)$ magnetic field. $\underline{E}$ are produced by charges and act upon charges. Similarly, $\underline{B}$ are produced by currents and act upon currents. In an electromagnetic medium, generally a non-uniform, characterised by "electric permittivity" $\varepsilon(\underline{x})$ and "magnetic permittivity" $\mu(\underline{x})$, in presence of electric charges and currents density $\rho(\underline{x}, t)$. Electric current density $\underline{j}(\underline{x}, t)$. The EM are described by Maxwell's equations (ME):

$$\nabla \cdot (\varepsilon \underline{E}) = \rho \quad (2)$$

$$\nabla \cdot \underline{B} = 0 \quad (3)$$

$$\nabla \times \underline{E} = -\frac{\partial \underline{B}}{\partial t} \quad (4)$$

$$\nabla \times \left(\frac{\underline{B}}{\mu} \right) = \varepsilon \frac{\partial \underline{E}}{\partial t} + \underline{j} \quad (5)$$

Remark.

- (1) MEs unify (for a uniform EM medium) generalisations to "densities" etc of The (1) refers to Coulomb's Law (in Gauss form). (2) refers to Gauss law for magnetic fields. (3) refers to Faraday's law of EM induction. (4) is Ampere's Law in Maxwell's form.
- (2) MEs implies conservation of electric charges.

$$\begin{aligned} \nabla \cdot \nabla \times \left(\frac{\underline{B}}{\mu} \right) &= \nabla \cdot \left(\varepsilon \frac{\partial \underline{E}}{\partial t} \right) + \nabla \cdot \underline{j} \\ \implies 0 &= \frac{\partial}{\partial t} (\nabla \cdot (\varepsilon \underline{E})) + \nabla \cdot \underline{j} \\ \implies \frac{\partial \rho}{\partial t} &= -\nabla \cdot \underline{j} \end{aligned}$$

In particular, for uniform (ε, μ constant). And source-free ($\rho \equiv 0, \underline{j} \equiv 0$). Maxwell's equations simplify to

$$\nabla \cdot \underline{E} = 0$$

$$\nabla \cdot \underline{B} = 0$$

$$\nabla \times \underline{E} = -\frac{\partial \underline{B}}{\partial t}$$

$$\nabla \times \underline{B} = \varepsilon \mu \frac{\partial \underline{E}}{\partial t}$$

To eliminate $\underline{B}$, take curl of $\nabla \times \underline{E} = -\frac{\partial \underline{B}}{\partial t}$ then we have

$$\begin{aligned} \nabla \times (\nabla \times \underline{E}) &= -\frac{\partial}{\partial t} (\nabla \times \underline{B}) \\ &= -\varepsilon \mu \frac{\partial^2 \underline{E}}{\partial t^2} \end{aligned}$$

Now, use vector identity we have

$$\nabla \times (\nabla \times \underline{E}) = \nabla(\nabla \cdot \underline{E}) - \nabla^2 \underline{E}$$

Thus,

$$\begin{aligned} \nabla(\nabla \cdot \underline{E}) - \nabla^2 \underline{E} &= -\varepsilon \mu \frac{\partial^2 \underline{E}}{\partial t^2} \\ \implies \nabla^2 \underline{E} &= \varepsilon \mu \frac{\partial^2 \underline{E}}{\partial t^2} \end{aligned}$$

then

$$\frac{1}{c^2} \frac{\partial^2 \underline{E}}{\partial t^2} - \nabla^2 \underline{E} = 0, \quad c := (\varepsilon \mu)^{-1/2}$$

By the same argument,

$$\frac{1}{c^2} \frac{\partial^2 \underline{B}}{\partial t^2} - \nabla^2 \underline{B} = 0$$

i.e. both $\underline{E}$ and $\underline{B}$ solves a vector wave equation, $c := (\mu \varepsilon)^{-1/2}$ is wavespeed. Physically, $c = c_0 \approx 3 \times 10^8 \text{ms}^{-1}$ in

vacuum. In any other EM medium, it can only be smaller.

In contrast to acoustic waves, not every solution will correspond to $\underline{E}$ or $\underline{B}$. However, if $\underline{E}$ solves the wave equation and $\nabla \cdot \underline{E} = 0$. Typically, there exists $\underline{B}$ such that $(\underline{E}, \underline{B})$ solve the MEs.

Similarly to acoustic, admits Time-Harmonic (T-H) solutions:

$$\begin{aligned}\underline{E}(\underline{x}, t) &= \text{Re}\{\hat{\underline{E}}(\underline{x})e^{-i\omega t}\} \\ \underline{B}(\underline{x}, t) &= \text{Re}\{\hat{\underline{B}}(\underline{x})e^{-i\omega t}\}\end{aligned}$$

where $\hat{\underline{E}}, \hat{\underline{B}}$ both solve vector Helmholtz equation:

$$\begin{aligned}\nabla^2 \hat{\underline{E}} + k^2 \hat{\underline{E}} &= 0 \\ \nabla^2 \hat{\underline{B}} + k^2 \hat{\underline{B}} &= 0\end{aligned}$$

where $k := \frac{\omega}{c} = \omega(\varepsilon\mu)^{1/2}$ is wavenumber.

Plane T-H EM waves: Similarly to acoustics, solutions of MEs admit plane T-H waves: Let WLOG z in propagation direction, we seek

$$\begin{aligned}\underline{E}(\underline{x}, t) &= \text{Re}\{\underline{E}_0 e^{-i\omega t + ikz}\} \\ \underline{B}(\underline{x}, t) &= \text{Re}\{\underline{B}_0 e^{-i\omega t + ikz}\}\end{aligned}$$

where for $k = \omega(\varepsilon\mu)^{1/2}$, these solve

$$\begin{aligned}\frac{1}{c^2} \frac{\partial^2 \underline{E}}{\partial t^2} - \nabla^2 \underline{E} &= 0, \quad c := (\varepsilon\mu)^{-1/2} \\ \frac{1}{c^2} \frac{\partial^2 \underline{B}}{\partial t^2} - \nabla^2 \underline{B} &= 0\end{aligned}$$

for any constant vectors $\underline{E}_0, \underline{B}_0 \in \mathbb{C}^3$. However, to solve MEs, $\underline{E}_0, \underline{B}_0$ must obey additional restrictions:

$$\nabla \cdot \underline{E} \implies \nabla \cdot (\text{Re}\{\underline{E}_0 e^{-i\omega t + ikz}\}) = 0 \text{ i.e.}$$

$$\text{Re}\{\nabla \cdot e^{-i\omega t + ikz}\} = \text{Re}\{E_z(ik)e^{-i\omega t + ikz}\} = 0$$

Thus, $E_z = 0$ so $\underline{E}_0 = (E_x, E_y, 0)$. Similarly, we have $\underline{B}_0 = (B_x, B_y, 0)$. So both (complex) $\underline{E}_0$ and $\underline{B}_0$ are orthogonal to wavevector $\underline{k} = k(0, 0, 1)$ of wave propagation: $\underline{E}_0 \cdot \underline{k} = \underline{B}_0 \cdot \underline{k} = 0$.

Next, let $\nabla \times \underline{E} = -\frac{\partial \underline{B}}{\partial t}$. Then

$$\begin{aligned}\nabla \times \underline{E} &= \nabla \times \text{Re}\{\underline{E}_0 e^{ikz - i\omega t}\} = \text{Re}\{e^{-i\omega t}(\nabla \times \underline{E}_0 e^{ikz})\} = \text{Re}\{e^{-i\omega t}(-ike^{ikz} E_y, ike^{ikz} E_x, 0)\} \\ &= \text{Re}\{ike^{ikz - i\omega t}(-E_y, E_x, 0)\} = -\frac{\partial \underline{B}}{\partial t} = \text{Re}\{\underline{B}_0(i\omega)e^{ikz - i\omega t}\}\end{aligned}$$

for all t and z . Thus, $\underline{B}_0 = (B_x, B_y, 0) = \frac{k}{\omega}(-E_y, E_x, 0) = \frac{1}{\omega}\underline{k} \times \underline{E}_0$. So $\underline{E}_0, \underline{B}_0$ and $\underline{k}$ are pairwise orthogonal, forming a right-handed triple. Note that $\underline{E}(\underline{x}, t) \perp \underline{k} \perp \underline{B}(\underline{x}, t) \perp \underline{E}(\underline{x}, t)$:

$$\underline{E}(\underline{x}, t) \cdot \underline{k} = \text{Re}\{\underline{E}_0 e^{ikz - i\omega t} \cdot \underline{k}\} = 0$$

we can repeat a similar argument for others. Also,

$$\underline{B}(\underline{x}, t) = \text{Re}\{\frac{1}{\omega}\underline{k} \times \underline{E}_0 e^{ikz - i\omega t}\} = \frac{1}{\omega}\underline{k} \times \text{Re}\{\underline{E}_0 e^{ikz - i\omega t}\} = \frac{1}{\omega}\underline{k} \times \underline{E}(\underline{x}, t)$$

So "true" $\underline{E}, \underline{B}$ and $\underline{k}$ remain (right-handed) $\perp$ triple. Polarisation plane P is $(\forall x, \forall t)$ is the plane formed by $\underline{E}_0(\underline{x}, t)$ and $\underline{k}$. Generally, it can rotate about $\underline{k}$ -axis with t and $\underline{x}$ since

$$\underline{E}(\underline{x}, t) = \text{Re}\{\underline{E}_0 e^{ikz - i\omega t}\} = (\text{Re } \underline{E}_0) \cos(\omega t - kz) + (\text{Im } \underline{E}_0) \sin(\omega t - kz)$$

So if $\text{Re } \underline{E}_0$ not collinear to $\text{Im } \underline{E}_0$, $\underline{E}(\underline{x}, t)$ will rotate and so will P . In this case, $\underline{E}(\underline{x}, t)$ traces out an ellipse in plane perpendicular to $\underline{k}$. However, if $(\text{Re } \underline{E}_0)$ and $(\text{Im } \underline{E}_0)$ are collinear, then polarisation plane P remains unchanged (or polarised).

Generally, any plane T-H wave is a sum of two polarised waves namely:

$$(\text{Re } \underline{E}_0) \cos(\omega t - kz) \text{ and } (\text{Im } \underline{E}_0) \sin(\omega t - kz)$$

with wavelength $\lambda = \frac{2\pi}{k} = \frac{2\pi}{\omega}c$, frequency $f = \frac{1}{T} = \frac{\omega}{2\pi}$, $T = \frac{2\pi}{\omega}$ is time-period. $\lambda f = c$ (as in acoustics). Depending on f , EM waves are found in different parts of EM "spectrum".

Poynting's vector: Similarly to acoustic, we can derive from MEs, the EM energy conservation law:

$$\begin{aligned}\underline{B} \cdot \nabla \times \underline{E} &= -\underline{B} \cdot \frac{\partial \underline{B}}{\partial t}, \quad \underline{E} \cdot \nabla \times \underline{B} = \varepsilon\mu \underline{E} \cdot \frac{\partial \underline{E}}{\partial t} \\ \underline{B} \cdot (\nabla \times \underline{E}) - \underline{E} \cdot (\nabla \times \underline{B}) &= -\underline{B} \cdot \frac{\partial \underline{B}}{\partial t} - \varepsilon\mu \underline{E} \cdot \frac{\partial \underline{E}}{\partial t}\end{aligned}$$

Recall a vector identity: for all $\underline{A}, \underline{B}$, $\nabla \cdot (\underline{A} \times \underline{B}) = (\nabla \times \underline{A}) \cdot \underline{B} - (\nabla \times \underline{B}) \cdot \underline{A}$. Therefore,

$$\begin{aligned}\nabla \cdot (\underline{E} \times \underline{B}) &= -\frac{\partial}{\partial t}(|\underline{B}|^2 + \varepsilon\mu \frac{|\underline{E}|^2}{2}) \\ \implies \frac{\partial}{\partial t}(\frac{\varepsilon}{2}|\underline{E}|^2 + \frac{1}{2\mu}|\underline{B}|^2) + \nabla \cdot (\frac{\underline{E} \times \underline{B}}{\mu}) &= 0\end{aligned}$$

which are exactly electric energy density and magnetic energy density, EM-flux.

$$\frac{\partial w}{\partial t} + \nabla \cdot \underline{S} = 0$$

where $w := \frac{\epsilon}{2} |\underline{E}|^2 + \frac{|B|^2}{2\mu}$ is EM energy density. $\underline{S} := \frac{\underline{E} \times \underline{B}}{\mu}$ is Poynting's vector described EM energy flow in direction of $\underline{S}$.

Example. For plane T-H polarised EM wave, e.g.

$$\begin{aligned} \underline{E}(x, t) &= (E_x, 0, 0) \cos(\omega t - kx), \quad E_x \in \mathbb{R} \\ \underline{B} &= \frac{1}{\omega} \underline{k} \times \underline{E} = \frac{1}{\omega} \begin{pmatrix} i & j & k \\ 0 & 0 & k \\ E_x & 0 & 0 \end{pmatrix} \cos(\omega t - kx) = \frac{1}{\omega} (0, kE_x, 0) \cos(\omega t - kx) = \frac{1}{c} (0, E_x, 0) \cos(\omega t - kx). \\ \underline{S} &:= \frac{1}{\mu} \underline{E} \times \underline{B} = \frac{1}{c\mu} \begin{pmatrix} i & j & k \\ E_x & 0 & 0 \\ 0 & E_x & 0 \end{pmatrix} \cos^2(\omega t - kx) = \frac{E_x^2}{\mu c} (0, 0, 1) \cos^2(\omega t - kx) \end{aligned}$$

So, as expected, EM energy flows in +ve z -direction (= direction of wave propagation).

3 Boundary and Interface Conditions

Both in Acoustics and EM, the equations have to be supplemented by appropriate boundary conditions (BC) or Interface Conditions (IC): Mathematically, typical BCs are:

- (1) Dirichlet BC: Unknown function $\Psi(x)$ sought in domain Ω bounded by surface S is assumed known on S .
- (2) Neumann BC: normal derivative of Ψ is prescribed on S : $\frac{\partial \Psi}{\partial n} := \underline{n} \cdot \nabla \Psi$. $\underline{n}$ unit normal exterior to S .
- (3) Mixed/Robin/Impedance BCs: a linear combination of Ψ and $\frac{\partial \Psi}{\partial n}$ known on S : $\alpha \Psi + \beta \frac{\partial \Psi}{\partial n}$ ($\alpha \beta \neq 0$) known.

3.1 Acoustic BCs

- (1) (Acoustically hard BCs) Let S be rigid, kinematically, acoustic velocity $\underline{u}$ must be parallel to S .

$$0 = \underline{u} \cdot \underline{n} = \nabla \phi \cdot \underline{n} =: \frac{\partial \phi}{\partial n}$$

which is exactly the Neumann BC.

- (2) (Acoustic soft BCs) In a "freely moving" surface, physically, the acoustic pressure must be zero:

$$p = 0 \text{ on } S$$

which is the Dirichlet BC for pressure.

- (3) (Impedance BC) Combining (1) and (2). The surface response is assumed proportional to p , and generally depends on frequency ω , being characterised by an "impedance" $Z = Z(\omega)$:

$$Z(\omega) := \frac{\hat{p}}{\hat{u}_n} \text{ where } \hat{u}_n := \hat{\underline{u}} \cdot \underline{n} \text{ "normal velocity" of } S$$

$$\hat{p} - Z \hat{u}_n = 0 \implies \hat{p} - Z \frac{\partial \hat{\phi}}{\partial n} = 0$$

Generally, $Z(\omega)$ is complex. And

$$\begin{aligned} \hat{p} &= i\rho_0 \omega \hat{\phi} \\ \implies i\rho_0 \omega \hat{\phi} - Z \frac{\partial \hat{\phi}}{\partial n} &= 0 \\ i\rho_0 \omega \hat{p} - Z \frac{\partial \hat{p}}{\partial n} &= 0 \end{aligned}$$

SO the Impedance BCs are mixed/Robin BC.

Remark.

- (1) As $Z \rightarrow 0$, we have $\hat{p} = 0$ equivalent to acoustically soft BC. If $Z \rightarrow \infty$, we have

$$\frac{\partial \hat{p}}{\partial n} = 0$$

equivalent to acoustically hard BC.

- (2) Physically, $\text{Re } Z \geq 0$ for energetically "neutral" ($\text{Re } Z = 0$) or "absorbing" ($\text{Re } Z > 0$).

3.2 EM Interface/Boundary Conditions

At an interface S between two EM media, with unit normal $\underline{n}$, the following ICs hold (from physics):

- (1) "Tangential" components of $\underline{E}$ are continuous (cts) across S :

$$\underline{E}^{(1)} \times \underline{n} = \underline{E}^{(2)} \times \underline{n}$$

- (2) "Normal" components of $\underline{B}$ are cts:

$$\underline{B}^{(1)} \cdot \underline{n} = \underline{B}^{(2)} \cdot \underline{n}$$

Also, generally, for a surface charge density ρ_s and surface current density $\underline{j}_s$.

$$(3) \quad \varepsilon_2 \underline{E}^{(2)} \cdot \underline{n} - \varepsilon_1 \underline{E}^{(1)} \cdot \underline{n} = \rho_s.$$

$$\frac{1}{\mu_1} \underline{B}^{(1)} \times \underline{n} - \frac{1}{\mu_2} \underline{B}^{(2)} \times \underline{n} = \underline{j}_s$$

In particular, if both media are insulators then $\rho_0 \equiv 0$ and $\underline{j}_s \equiv 0$. Then normal component of $\varepsilon \underline{E}$ and tangential components of $\underline{B}/\mu$ are cts.

Remark. TC can be obtained from the inhomogeneous MEs as a limit when $\varepsilon(\underline{x}), \mu(\underline{x})$.

Perfectly Conducting (PC) BCs: Let medium 2 be a PC then $\underline{E}^{(2)} \equiv 0$ physically. For $\underline{E}^{(1)} \equiv \underline{E}$ we have $\underline{E} \times \underline{n} = 0$ which is the PC BC. $\underline{E}$ has only normal component on PC surface S .

4 Canonical Cases

4.1 Reflection by a plane (acoustics)

Let a plane T-H acoustic wave be incident upon a plane boundary

$$S = \{\underline{x} = (x, y, z) \mid x = 0\}$$

with angle of incidence Θ_I with $0 \leq \Theta_I < \frac{\pi}{2}$ towards S : $p_I(\underline{x}, t) = A_I e^{i \underline{k}_I \cdot \underline{x} - i \omega t}$ where $\underline{k}_I = k(\cos \theta_I, 0, \sin \theta_I)$ so

$$p_I = A_I e^{i k \cos \theta_I x + i k \sin \theta_I z - i \omega t}$$

$\omega > 0$, $k = \omega/c$, $A_I \in \mathbb{C}$. S reflects p_I , producing a reflected wave P_R , which we seek to be also plane T-H wave:

$$P_R = A_R e^{i \underline{k}_R \cdot \underline{x} - i \omega_R t}$$

with unknowns: $\omega_R > 0$, $\underline{k}_R = (k_x, k_y, k_z)$, $A_R \in \mathbb{C}$. P_R will depend on BC on S . Assume a acoustically soft BC: $p = p_I + p_R = 0$ on S i.e. $P_R = -P_I$ for $x = 0$. Note, need to require $k_x < 0$ (P_R "away" from S). Now, the BC ($x = 0$) we have

$$A_R e^{i k_y y + i k_z z - i \omega_R t} = -A_I e^{i k_z \sin \theta_I z - i \omega t}, \quad \forall t, \forall z, \forall y$$

Varying t e.g. for $z = y = 0$ we get $\omega_R = \omega$. Similarly we have $A_R = -A_I$, $k_y = 0$, $k_z = k \sin \theta_I$. Now, $|\underline{k}_R| = k_R \frac{\omega_R}{c} = \frac{\omega}{c} = k$ so

$$|\underline{k}_R| = \sqrt{k_x^2 + k^2 \sin^2 \theta_I} = k \implies k_x = -k \cos \theta_I$$

So

$$P_R = -A_I e^{i \underline{k}_R \cdot \underline{x} - i \omega t}$$

where $\underline{k}_R = k(-\cos \theta_I, 0, \sin \theta_I)$. Interpretation: "Specular Reflection Law", angle of reflection equals to the angle of incidence.

Exercise.

(1) Acoustically hard BC implies P_R as above, except $A_R = A_I$.

(2) Impedance BC.

4.2 Reflection of EM waves by P.C. Plane

Plane T-H EM wave $(\underline{E}_I, \underline{B}_I)$ propagating in half space $\{x < 0\}$ in EM medium i.e. ε, μ is incident upon a Perfectly Conducting (PC) plane $S = \{x = 0\}$, $0 \leq \theta < \pi/2$ angle of incidence; Let $\underline{E}_I$ be polarised so that point out of plane:

$$\underline{k}_I = k_I(\cos \theta, 0, \sin \theta), k_I = \frac{\omega}{c}, c = (\varepsilon \mu)^{-1/2}, \quad \underline{E}_I = (0, E_I, 0) e^{-i \omega t + i k_I (x \cos \theta + z \sin \theta)}$$

$$\underline{B}_I = \frac{E_I}{c} (-\sin \theta, 0, \cos \theta) e^{-i \omega t + i k_I (x \cos \theta + z \sin \theta)}$$

Seek $(\underline{E}_R, \underline{B}_R)$ as plane T-H wave, with angle of reflection $0 \leq \theta_R < \pi/2$. Let

$$\underline{E}_R = (E_x, E_y, E_z) e^{-i \omega_R t + i k_R (-x \cos \theta_R + z \sin \theta_R)}$$

and

$$\underline{B}_R = (B_x, B_y, B_z) e^{-i \omega_R t + i k_R (-x \cos \theta_R + z \sin \theta_R)}$$

We shall use the result that $\underline{E} \times \underline{n} = 0$ where $\underline{E} = \underline{E}_I + \underline{E}_R$.

$$E_I e^{-i \omega t + i k_I z \sin \theta} = -E_R e^{-i \omega_R t + i k_R z \sin \theta_R}, \quad \forall t, \forall z$$

$$\implies \omega_R = \omega, \quad k_R = k_I = k, \quad \theta = \theta_R, \quad E_I = E_y$$

same as in acoustics. Similarly, we get $E_z = 0$. Also, for plane T-H EM waves,

$$\underline{E}_R \perp \underline{k}_R \implies \underline{E}_R \cdot \underline{k}_R = 0$$

$$-E_x \cos \theta + E_z \sin \theta = 0 \implies E_x = 0$$

So $\underline{E}_R = (0, -E_I, 0) e^{-i \omega t + i \underline{k}_R \cdot \underline{x}}$. So E_R is also point along y -axis known as the Transverse Electric (TE) field. Finally,

$$\underline{B}_R = \frac{1}{\omega} \underline{k}_R \times \underline{E}_R = \frac{k}{\omega} \begin{pmatrix} i & j & k \\ -\cos \theta & 0 & \sin \theta \\ 0 & -E_I & 0 \end{pmatrix} e^{-i \omega t + i \underline{k}_R \cdot \underline{x}} = \frac{E_I}{c} (\sin \theta, 0, \cos \theta) e^{-i \omega t + i \underline{k}_R \cdot \underline{x}}$$

Exercise. Similar analysis for Transverse Magnetic (TM) case.

Back to our example above, there are "induced current/charges" on a PC's surface S : From Interface Condition (IC)

$$\frac{1}{\mu_1} \underline{B}^{(1)} \times \underline{n} - \frac{1}{\mu_2} \underline{B}^{(2)} \times \underline{n} = \underline{j}_s$$

In our case:

$$\underline{B}^{(1)} = \underline{B}_I + \underline{B}_R.$$

$$\underline{E}^{(2)} = 0 \implies \frac{\partial \underline{B}^{(2)}}{\partial t} = -\nabla \times \underline{E}^{(2)} = 0 \implies \underline{B}^{(2)} = 0$$

Thus,

$$\underline{j}_s = \frac{1}{\mu} \underline{B}^{(1)} \times \underline{n} = \frac{1}{\mu} (\underline{B}_I + \underline{B}_R) \times \underline{n} \text{ on } S$$

$$\underline{n} = (1, 0, 0), \text{ now on } S, x = 0. \text{ Recall } \underline{B}_I = \frac{E_I}{c} (-\sin \theta, 0, \cos \theta) e^{-i\omega t + i\mathbf{k}_I \cdot \mathbf{x}} \text{ and}$$

$$\underline{B}_R = \frac{E_I}{c} (\sin \theta, 0, \cos \theta) e^{-i\omega t + i\mathbf{k}_I \cdot \mathbf{x}}$$

$$\implies \underline{B}_I + \underline{B}_R = \frac{E_I}{c} (0, 0, 2 \cos \theta) e^{-i\omega t + i\mathbf{k}_I \cdot \mathbf{x}}$$

So

$$\underline{j}_s = \frac{E_I}{c\mu} \begin{pmatrix} i & j & k \\ 0 & 0 & 2 \cos \theta \\ 1 & 0 & 0 \end{pmatrix} e^{-i\omega t + i\mathbf{k}_I \cdot \mathbf{x}} = \frac{2 \cos \theta E_I}{c\mu} (0, 1, 0) e^{-i\omega t + i\mathbf{k}_I \cdot \mathbf{x}}$$

4.3 Reflection and Refraction by a plane EM Interface

Plane T-H EM wave, TE-polarised is incident from Medium 1 ($\varepsilon_1, \mu_1, c_1 = (\varepsilon_1 \mu_1)^{-1/2}$) upon interface $S = \{x = 0\}$ with Medium 2 ($\varepsilon_2, \mu_2, c_2 = (\varepsilon_2 \mu_2)^{-1/2}$). Let $(\underline{E}_I, \underline{B}_I)$ be as before where

$$\underline{E}_I = (0, E_I, 0) e^{-i\omega t + i\mathbf{k}_I \cdot \mathbf{x}} \quad \underline{B}_I = \frac{E_I}{c} (-\sin \theta, 0, \cos \theta) e^{-i\omega t + i\mathbf{k}_I \cdot \mathbf{x}}$$

We shall seek $(\underline{E}_R, \underline{B}_R)$ with

$$\underline{E}_R = (E_x, E_y, E_z) e^{-i\omega_R t + i\mathbf{k}_R \cdot \mathbf{x}}, \quad \underline{k}_R = \frac{\omega_R}{c_1} (-\cos \theta_R, 0, \sin \theta_R)$$

$$\underline{B}_R = (B_x, B_y, B_z) e^{-i\omega_R t + i\mathbf{k}_R \cdot \mathbf{x}}$$

Additionally, seek transmitted plane T-H wave as:

$$\underline{E}_T = (\hat{E}_x, \hat{E}_y, \hat{E}_z) e^{-i\omega_T t + i\mathbf{k}_T \cdot \mathbf{x}}, \quad \underline{k}_T = \frac{\omega_T}{c_2} (\cos \theta_T, 0, \sin \theta_T)$$

$$\underline{B}_T = (\hat{B}_x, \hat{B}_y, \hat{B}_z) e^{-i\omega_T t + i\mathbf{k}_T \cdot \mathbf{x}}$$

All the unknowns have to be found from the EM Interface Conditions (IC).

$$\underline{E}^{(1)} \times \underline{n} = \underline{E}^{(2)} \times \underline{n}$$

i.e. y, z components of $\underline{E}$ continuous (cts). Note that

$$\underline{E}^{(1)} = \underline{E}_I + \underline{E}_R, \quad \underline{E}^{(2)} = \underline{E}_T \text{ on } S \text{ i.e. } x = 0$$

At y -component:

$$E_I e^{-i\omega t + i\mathbf{k}_I \cdot \mathbf{x}} + E_y e^{-i\omega_R t + i\mathbf{k}_R \cdot \mathbf{x}} = \hat{E}_y e^{-i\omega_T t + i\mathbf{k}_T \cdot \mathbf{x}}, \quad \forall t, \forall z$$

Now, varying t , and then varying z we get

$$\omega_R = \omega_T = \omega, \quad k \sin \theta = k_R \sin \theta_R = k_T \sin \theta_T$$

Now $k_R = \frac{\omega_R}{c_1} = \frac{\omega}{c_1} = k$ implies $\theta_R = \theta$ (Specular Reflection Law again). Also $k_T = \frac{\omega_T}{c_2} = \frac{\omega}{c_2}$ so we have

$$\frac{\omega}{c_1} \sin \theta = \frac{\omega}{c_2} \sin \theta_T$$

$$\implies \frac{\sin \theta_T}{c_2} = \frac{\sin \theta}{c_1}$$

Equivalently, introducing "refractive" index: $n := \frac{c_0}{c}$ where c is the wave speed in the EM medium and c_0 is the speed of light (= wavespeed in vacuum) i.e. $n_1 = \frac{c_0}{c_1} \geq 1, n_2 = \frac{c_0}{c_2} \geq 1$. (n is the EM medium's refractive index). Then we have

$$n_2 \sin \theta_T = n_1 \sin \theta$$

known as the Snell's Law of refraction. Also, we have $E_I + E_y = \hat{E}_y$. Now in z -component we have $E_z = \hat{E}_z$. Now, as $\underline{E}_R$ and $\underline{E}_T$ are plane T-H implies $\underline{E}_R \cdot \underline{k}_R = 0, \underline{E} \cdot \underline{k}_T = 0$.

$$-E_x \cos \theta + E_z \sin \theta = 0$$

$$\hat{E}_x \cos \theta_T + \hat{E}_z \sin \theta_T = 0$$

Next, $\varepsilon_1 \underline{E}^{(1)} \cdot \underline{n} - \varepsilon_2 \underline{E}^{(2)} \cdot \underline{n} = 0$. $\varepsilon_1 E_x = \varepsilon_2 \hat{E}_x$. Notice that

$$E_z = \hat{E}_z = -\hat{E}_x \cot \theta_T = -\frac{\varepsilon_1}{\varepsilon_2} \cot \theta_T E_x = -\frac{\varepsilon_1}{\varepsilon_2} \cot \theta_T \tan \theta E_z$$

$$\implies E_z \equiv 0$$

Therefore, $E_z = \hat{E}_z = \hat{E}_x = E_x = 0$. So as the incident wave, both $\underline{E}_R$ and $\underline{E}_T$ remain transverse Electric (TE). Need one more equation for E_I and $\hat{E}_y$, from the remaining interface conditions. Both involving $\underline{B}$; Now

$$\underline{B}_R = \frac{1}{\omega} \underline{k}_R \times \underline{E}_R = \frac{k}{\omega} \begin{pmatrix} i & j & k \\ -\cos \theta & 0 & \sin \theta \\ 0 & E_y & 0 \end{pmatrix} e^{i \underline{k}_R \cdot \underline{x} - i \omega t} = \frac{1}{c_1} (-E_y \sin \theta, 0, -E_y \cos \theta) e^{i \underline{k}_R \cdot \underline{x} - i \omega t}$$

Similarly,

$$\underline{B}_T = \frac{1}{\omega} \underline{k}_T \times \underline{E}_T = \frac{k_T}{\omega} \begin{pmatrix} i & j & k \\ \cos \theta_T & 0 & \sin \theta_T \\ 0 & \hat{E}_y & 0 \end{pmatrix} e^{i \underline{k}_T \cdot \underline{x} - i \omega t} = \frac{1}{c_2} (-\hat{E}_y \sin \theta_T, 0, \hat{E}_y \cos \theta_T) e^{i \underline{k}_T \cdot \underline{x} - i \omega t}$$

In addition we have

$$\underline{B}_T = \frac{1}{c_1} (-E_I \sin \theta, 0, E_I \cos \theta) e^{i \underline{k}_I \cdot \underline{x} - i \omega t}$$

We let $\underline{j}_s = 0$ for insulator media so we have $\frac{1}{\mu} \underline{B}$ has continuous tangential components. If we look at the z -component:

$$\frac{\cos \theta}{\mu_1 c_1} (E_I - E_y) = \frac{\cos \theta_T}{\mu_2 c_2} \hat{E}_y$$

Use $\hat{E}_y = E_I + E_y$ we have

$$m_1(E_I - E_y) = m_2(E_I + E_y) \\ E_y = \frac{m_1 - m_2}{m_1 + m_2} E_I, \quad \hat{E}_y = \frac{2m_1}{m_1 + m_2} E_I.$$

where

$$m_1 := \frac{\cos \theta}{\mu_1 c_1}, \quad m_2 := \frac{\cos \theta_T}{\mu_2 c_2}$$

In particular, if both media are non-magnetic i.e. $\mu_1 = \mu_2 = \mu$ then $(n_1 := \frac{c_0}{c_1}, n_2 := \frac{c_0}{c_2})$. Then

$$E_y = \frac{n_1 \cos \theta - n_2 \cos \theta_T}{n_1 \cos \theta + n_2 \cos \theta_T} E_I$$

and

$$\hat{E}_y = \frac{2n_1 \cos \theta}{n_1 \cos \theta + n_2 \cos \theta_T} E_I$$

Notice, $|E_y| \leq |E_I|$ as expected since Reflected amplitude less or equal to incident one.

4.4 Total Internal Reflection (TIR)

Recall Snell's refraction law

$$\frac{\sin \theta_T}{c_2} = \frac{\sin \theta}{c_1} \\ \sin \theta_T = \frac{c_2}{c_1} \sin \theta \leq 1 \Leftrightarrow \sin \theta \leq \frac{c_1}{c_2}$$

Now if $c_2 > c_1$ then $\theta \leq \arcsin\left(\frac{c_1}{c_2}\right) = \theta_c$ the critical angle. So if $c_2 > c_1$ and $\theta > \theta_c$. No real θ_T exists. However, there exists a complex θ_T known as the Total Internal Reflection. Now let $c_2 > c_1$ and $\theta > \theta_c$ then

$$\sin \theta_T = \frac{c_2}{c_1} \sin \theta =: \alpha > 1, \quad \cos \theta_T = (1 - \sin^2 \theta_T)^{1/2} = i\sqrt{\alpha^2 - 1} = i\beta, \beta > 0$$

Recall that

$$E_y = \frac{n_1 \cos \theta - n_2 \cos \theta_T}{n_1 \cos \theta + n_2 \cos \theta_T} E_I = \frac{n_1 \cos \theta - i\beta n_2}{n_1 \cos \theta + i\beta n_2} E_I \\ \Rightarrow |E_y| = |E_I|$$

so reflected energy equals to incident one so energy is reflected back. What happens in medium 2 under TIR? ($\theta > \theta_c$).

$$\underline{E}_T = \text{Re}\{(0, \hat{E}_y, 0) e^{i \underline{k}_T \cdot \underline{x} - i \omega t}\}$$

where $\underline{k}_T = k_T(\cos \theta_T, 0, \sin \theta_T) = k_T(i\beta, 0, \alpha)$. Thus, $i \underline{k}_T \cdot \underline{x} = -\beta k_T x + i \alpha k_T z$. So

$$\underline{E}_T = \text{Re}\{(0, \hat{E}_y, 0) e^{-\beta k_T x + i \alpha k_T z - i \omega t}\} = (0, |\hat{E}_y| e^{-\beta k_T x} \cos(\omega t - \alpha k_T z - \psi), 0)$$

where $\hat{E}_y = |\hat{E}_y| e^{i\psi}$. So as x increases, $\underline{E}_T$ decays exponentially. Let $\varphi := \omega t - \alpha k_T z - \psi$. Next,

$$\underline{B}_T(x, t) = \text{Re}\{\underline{B}_0^T e^{-i \omega t + i \underline{k}_T \cdot \underline{x}}\}$$

where $\underline{B}_0^T = \frac{1}{\omega} \underline{k}_T \times \underline{E}_0^T$, $\underline{E}_0^T = (0, \hat{E}_y, 0)$.

$$\underline{B}_0^T = \frac{k_T}{\omega} \begin{pmatrix} i & j & k \\ i\beta & 0 & \alpha \\ 0 & \hat{E}_y & 0 \end{pmatrix} = \frac{1}{c_2} (-\alpha \hat{E}_y, 0, i\beta \hat{E}_y)$$

$$\underline{B}_T = \text{Re}\left\{ \frac{\hat{E}_y}{c_2} (-\alpha, 0, i\beta) e^{-i \omega t - \beta k_T x + i \alpha k_T z} \right\} \\ = \frac{|\hat{E}_y|}{c_2} e^{-\beta k_T x} (-\alpha \cos \varphi, 0, \beta \sin \varphi)$$

Now, the Poynting's vector

$$\underline{S}_T := \frac{1}{\mu_2} \underline{E}_T \times \underline{B}_T = \frac{|\hat{E}_y|^2 e^{-2\beta k_T x}}{\mu_2 c_2} \begin{pmatrix} i & j & k \\ 0 & \cos \varphi & 0 \\ -\alpha \cos \varphi & 0 & \beta \sin \varphi \end{pmatrix} = \frac{|\hat{E}_y|^2 e^{-2\beta k_T x}}{\mu_2 c_2} (\beta \sin \varphi \cos \varphi, 0, \alpha \cos^2 \varphi)$$

So, for time-averaged $\langle \underline{S}_T \rangle$ as $\langle \cos^2 \varphi \rangle = \frac{1}{2}$ and $\langle \sin \varphi \cos \varphi \rangle = 0$. Thus,

$$\langle \underline{S}_T \rangle = \frac{|\hat{E}_y|^2 e^{-2\beta k_T x} \alpha}{2\mu_2 c_2} (0, 0, 1)$$

Interpretation: The EM in $\{x > 0\}$ decays exponentially and time-averaged EM energy is transported along interface (in +ve z-direction).

Remark.

1. Similar analysis for "Transverse Magnetic" (TM) waves. Can it happen that there is no reflection? Equivalently, $E_y = 0$ i.e.

$$E_y = \frac{n_1 \cos \theta - n_2 \cos \theta_T}{n_1 \cos \theta + n_2 \cos \theta_T} E_I = 0$$

$\implies n_1 \cos \theta = n_2 \cos \theta_T$ but from Snell's Law we have $n_1 \sin \theta = n_2 \sin \theta_T$ which means $\tan \theta = \tan \theta_T$. Thus, $\theta_T = \theta$ and $n_1 = n_2$. This suggests that two media are identical. So always reflection for two different non-magnetic media in TE polarisation. However, I love my baby. for TM ($\mu_1 = \mu_2$), there exists an angle Brewster's angle of incidence θ_B such that $E_y = 0$.

2. Any polarisation of $\underline{E}_I$ is a sum of TM and TE waves. Since TE and TM waves have different reflective/transmissive properties. If $\theta \approx \theta_B$. Only TE polarisation is reflected.
3. In many materials, $\varepsilon = \varepsilon(\omega)$ varies with frequency of EM waves. By Snell's Law, $\theta_T = \theta_T(\omega)$.
4. Similar refraction+reflection at an acoustic interface.
5. I love my baby.

4.5 Rapidly Pulsating Spherical Sources

An acoustic medium is perturbed by a rapidly pulsating sphere: its radius $a = a(t)$ varies around a reference value a_0 , $|a(t) - a_0| \ll a_0$ i.e. small perturbations. Then

$$\underline{V}_s(t) = V_s(t) \tilde{r}, \quad V_s(t) = \frac{da}{dt} = a'(t).$$

radical velocity $\tilde{r} := \frac{\underline{x}}{|\underline{x}|} = \frac{\underline{x}}{r}$. Unit vector $\underline{u}$ outer radical direction. Assume the resulting acoustic field spherical-symmetric outgoing (casual) and irrotational. Therefore, $\underline{u} = \nabla \phi$ where $\phi(r, t)$, $r \equiv |\underline{x}|$. Then, compare to $\phi(r, t) = \frac{f(r-ct)}{r}$. Hence, $\underline{u} = \nabla \phi = \frac{\partial \phi}{\partial r} \tilde{r}$. BCs: at $r = a_0$, $\underline{u} = \underline{V}_s$ so

$$\begin{aligned} \frac{\partial \phi}{\partial r} \tilde{r} &= V_s(t) \tilde{r} \\ \frac{\partial \phi}{\partial r} &= V_s(t), \text{ on } r = a_0 \end{aligned}$$

To find $f(\xi)$ equivalently seek

$$\phi(r, t) = \frac{f(r-ct)}{r} = -ca_0 \frac{\psi(t - \frac{r}{c} + \frac{a_0}{c})}{r}$$

Now,

$$\begin{aligned} \frac{\partial \phi}{\partial r} &= \frac{ca_0}{r^2} \psi(t - \frac{r}{c} + \frac{a_0}{c}) - ca_0 \left(-\frac{1}{c}\right) \frac{\psi'(t - \frac{r}{c} + \frac{a_0}{c})}{r} \\ \frac{\partial \phi}{\partial r} &= \psi'(t) + \frac{c}{a_0} \psi(t) = V_s(t) \end{aligned}$$

which is an ODE for $\psi(t)$. For an initial condition, assuming for all $t \leq t_0$, $\psi(t) \equiv 0$ implies $\psi(t_0) = 0$. Using integrating factor we get

$$\begin{aligned} \frac{d}{dt} (e^{\frac{ct}{a_0}} \psi(t)) &= e^{\frac{ct}{a_0}} V_s(t) \\ \implies e^{\frac{ct}{a_0}} \psi(t) &= \int_{t_0}^t e^{\frac{c\tau}{a_0}} V_s(\tau) d\tau \end{aligned}$$

Thus,

$$\psi(t) = \int_{t_0}^t e^{\frac{c(\tau-t)}{a_0}} V_s(\tau) d\tau$$

is the solution. Since $u = \nabla \phi$ and $p = -\rho_0 \frac{\partial \phi}{\partial t}$. For T-H case, $V_s(t) = \text{Re}\{B e^{-i\omega t}\}$, $B \in \mathbb{C}$. And seek ψ also in the T-H form:

$$\psi(t) = \text{Re}\{B e^{-i\omega t}\}, B \in \mathbb{C}$$

to be found. Then

$$\begin{aligned} \text{Re}\{(-i\omega)B e^{-i\omega t} + \frac{c}{a_0} B e^{-i\omega t}\} &= \text{Re}\{\hat{V}_s e^{-i\omega t}\} \quad \forall t \\ \implies B &= \frac{\hat{V}_s}{\frac{c}{a_0} - i\omega} = \frac{a_0 \hat{V}_s}{c - i\omega a_0} = \frac{a_0 \hat{V}_s}{c(1 - ika_0)} \end{aligned}$$

$$\begin{aligned}\phi &= -\frac{ca_0}{r}\psi\left(t - \frac{r}{c} + \frac{a_0}{c}\right) = -\frac{ca_0}{r}\operatorname{Re}\{Be^{-i\omega(t - \frac{r}{c} + \frac{a_0}{c})}\} \\ &= \operatorname{Re}\left\{-\frac{ca_0}{r}\frac{a_0\hat{V}_s}{c(1 - ika_0)}e^{-i\omega t + ikr - ika_0}\right\} = \operatorname{Re}\left\{-\frac{a_0^2\hat{V}_s}{(1 - ika_0)r}e^{ikr - ika_0}e^{-i\omega t}\right\}\end{aligned}$$

Since $\hat{p} = i\omega\rho_0\hat{\phi}$. Then

$$\begin{aligned}p &= \operatorname{Re}\left\{-\frac{i\omega\rho_0a_0^2\hat{V}_s}{1 - ika_0}e^{-ika_0}\frac{e^{ikr}}{r}e^{-i\omega t}\right\} \\ \implies p &= \operatorname{Re}\left\{A\frac{e^{ikr}}{r}e^{-i\omega t}\right\}\end{aligned}$$

where

$$A := -\frac{i\omega\rho_0a_0^2\hat{V}_s}{1 - ika_0}e^{-ika_0}$$

5 Green's functions, Green's formulas

5.1 Points Sources

As seen before, any T-H acoustic field due to radially pulsing are of radius $a_0 > 0$, of angular frequency $\omega > 0$ is of the form $p = \operatorname{Re}\{\hat{p}e^{-i\omega t}\}$ where $\hat{p} = A\frac{e^{ikr}}{r}$, $r = |\underline{x}|$ and

$$A := -\frac{i\omega\rho_0a_0^2\hat{V}_s}{1 - ika_0}e^{-ika_0} \in \mathbb{C}$$

Fix $A \neq 0$, and let $a_0 \rightarrow 0$ changing $\hat{V}_s$ accordingly, the "limit" holds for $r \neq 0$ can be viewed as a field due to "point source" at $\underline{x} = 0$. Recall for $r \neq 0$, $\hat{p}$ solves the Helmholtz equation:

$$(\nabla^2\hat{p} + k^2\hat{p}) = 0, \quad r \neq 0$$

Can we make sense of the above equation for all r (including $r = 0$). For small $\varepsilon > 0$, let $\hat{p}_\varepsilon = \hat{p}$ for all $r \geq \varepsilon$. But for $0 \leq r < \varepsilon$, $\hat{p}_\varepsilon$ is smooth. Then,

$$\Delta^2\hat{p}_\varepsilon + k^2\hat{p}_\varepsilon = \Delta_\varepsilon(r)$$

$\Delta_\varepsilon(r)$ can be compactly supported on B_ε ball of radius ε centred at zero. Let $\varepsilon \rightarrow 0$, and interpret over a ball B_R with $R > 0$. Then

$$\int_{B_R} \Delta\hat{p}_\varepsilon dV + \int_{B_R} k^2\hat{p}_\varepsilon dV = \int_{B_R} \Delta_\varepsilon dV$$

Next,

$$\begin{aligned}\int_{B_R} \Delta\hat{p}_\varepsilon dV &= \int_{B_R} \nabla \cdot (\nabla\hat{p}_\varepsilon) dV = \int_{S_R} \nabla\hat{p} \cdot \underline{n} dS = \int_{S_R} \frac{\partial\hat{p}}{\partial r} dS \\ &= \int_{S_R} A\left(-\frac{e^{ikr}}{r^2} + \frac{ike^{ikr}}{r}\right) dS = A\left(\frac{ik}{R} - \frac{1}{R^2}\right)e^{ikR} \times 4\pi R^2 = 4\pi A(ikR - 1)e^{ikR}\end{aligned}$$

Next, take $\varepsilon \rightarrow 0$

$$\begin{aligned}\int_{B_R} k^2\hat{p}_\varepsilon dV &\rightarrow \int_{B_R} k^2\hat{p} dV = \int_0^{2\pi} \int_0^\pi \int_0^R k^2 A \frac{e^{ikr}}{r} r^2 \sin\theta dr d\theta d\varphi \\ &= Ak^2 \left(\int_0^{2\pi} \int_0^\pi \sin\theta d\theta d\varphi\right) \int_0^R e^{ikr} r dr = 4\pi A e^{ikR}(1 - ikR) - 4\pi A\end{aligned}$$

Thus, as $\varepsilon \rightarrow 0$

$$\begin{aligned}\int_{B_R} \Delta_\varepsilon dV &\rightarrow 4\pi A(ikR - 1)e^{ikR} \rightarrow -4\pi A \\ &\quad \int_{B_R} \Delta_\varepsilon\end{aligned}$$

Let $\delta_\varepsilon(\underline{x}) = -\frac{1}{4\pi A}\Delta_\varepsilon$ then observe

$$\int_{B_R} \delta_\varepsilon(\underline{x}) dV \rightarrow 1 \text{ as } \varepsilon \rightarrow 0$$

and δ_ε is compactly supported in B_ε . Such δ_ε are called "delta sequences" and "limit" $\delta_\varepsilon(\underline{x}) \rightarrow \delta(\underline{x})$ where the "appropriate" limit, $\delta(\underline{x})$ is called Dirac δ -function. So $\delta(\underline{x})$ can be viewed as a limit "function":

$$\delta(\underline{x}) = \begin{cases} 0, & \underline{x} \neq 0 \\ +\infty & \underline{x} = 0 \end{cases}$$

$$\int_V \delta(\underline{x}) dV = 1, \quad \forall V \ni 0$$

Rigorous definition via distribution theory. One property of $\delta(\underline{x})$, "shifting property": for all "reasonable" $f(\underline{x})$ e.g. continuous for all $\underline{x} \in V$.

$$\int \delta(\underline{x} - \underline{x}') f(\underline{x}') dV(\underline{x}') = \int \delta(\underline{x}' - \underline{x}) f(\underline{x}') dV(\underline{x}') = f(\underline{x})$$

If we go back,

$$\hat{p} = A \frac{e^{ikr}}{r}$$

$$(\nabla^2 + k^2)\hat{p}_\varepsilon = \Delta_\varepsilon = -4\pi A\delta_\varepsilon$$

Take $\varepsilon \rightarrow 0$ we have

$$(\nabla^2 + k^2)\hat{p} = -4\pi A\delta(\underline{x})$$

More generally, for points sources as $\underline{x}_s$,

$$\hat{p}(\underline{x}) = A \frac{e^{ikR}}{R} \quad R := |\underline{x} - \underline{x}_s|$$

Green's functions: solutions of the above equation with $A = -\frac{1}{4\pi}$ in whole space with no boundaries is called free-space Green's function. $G(\underline{x}, \underline{x}') = G_f(\underline{x}, \underline{x}')$ then

$$(\nabla^2 + k^2)G(\underline{x}, \underline{x}') = \delta(\underline{x} - \underline{x}')$$

is the "point-source" at $\underline{x}'$. Hence,

$$G_f(\underline{x}, \underline{x}') = -\frac{e^{ik|\underline{x} - \underline{x}'|}}{4\pi|\underline{x} - \underline{x}'|}.$$

5.2 Green's Functions

Consider the field due to an arbitrary 'distributed' source of density $f(x)$:

$$\mathcal{L}\hat{p} := (\nabla^2 + k^2)\hat{p} = f(x),$$

where $f(x)$ is continuous and compactly supported. $\mathcal{L}$ is a (Helmholtz) differential operator. We could form a "well-posed" boundary value problem given PDEs+BCs+SRC. There exists unique solution $\hat{p} : f \mapsto \hat{p} = \mathcal{L}^{-1}f$. Try seeking $\mathcal{L}^{-1}$ as an (linear) integral operator:

$$\hat{p}(x) = (\mathcal{L}^{-1}f)(\underline{x}) = \int_{\Omega} K(\underline{x}, \underline{x}')f(\underline{x}')d\underline{x}'$$

with "kernel" $K(\underline{x}, \underline{x}')$.

$$\begin{aligned} f(x) &= \mathcal{L}\hat{p} = \mathcal{L}\mathcal{L}^{-1}f = \mathcal{L} \int_{\Omega} K(\underline{x}, \underline{x}')f(\underline{x}')d\underline{x}' \\ &= \int_{\Omega} \mathcal{L}_x K(\underline{x}, \underline{x}')f(\underline{x}')d\underline{x}' \end{aligned}$$

Notice that if $\mathcal{L}_x K(\underline{x}, \underline{x}') \equiv \delta(\underline{x} - \underline{x}')$ this serves the purposes. Now, we have

$$K(\underline{x}, \underline{x}') = G(\underline{x}, \underline{x}')$$

For domains Ω with boundaries GFs are generally in form"

$$G(\underline{x}, \underline{x}') = G_f(\underline{x}, \underline{x}') + G_B(\underline{x}, \underline{x}')$$

where f is free-space and G_B solves

$$(\nabla^2 + k^2)G_B = 0 \text{ in } \Omega$$

and $G_f + G_B$ obeys the BCs.

5.3 Kirchhoff-Helmholz Integral Theorem

That's a representative for (acoustic) field outside scattering boundaries. It is based on Green's identity: For $f, g \in C^2$ in V bounded by S with unit outer normal $\underline{n}$, apply Divergence theorem for $\underline{F} = f\nabla g - g\nabla f$ we have

$$\begin{aligned} \int_V \nabla \cdot \underline{F} dV &= \int_S \underline{F} \cdot \underline{n} dS : \\ \int_V \nabla f \nabla^2 g - g \nabla^2 f dV &= \int_S f \frac{\partial g}{\partial n} - g \frac{\partial f}{\partial n} dS \end{aligned}$$

which is the Green's identity. Let ψ be a solution to Helmholtz equation i.e. $(\nabla^2 + k^2)\psi = 0$ in domain Ω and bounded by S_0 . Let $f(x) = \psi(x)$, $g(\underline{x}) = G(\underline{x}, \underline{x}')$, $\underline{x} \in \Omega$ and $V = V_R$ is part of Ω inside ball B_R of large enough radius R centred at 0. Then we have

$$\int_{V_R} [\psi(\underline{x}) \nabla_x^2 G(\underline{x}, \underline{x}') - G(\underline{x}, \underline{x}') \nabla^2 \psi(\underline{x})] dV = \int_{V_R} \psi(\underline{x}) \delta(\underline{x} - \underline{x}') dV = \psi(\underline{x}')$$

Note that

$$\psi(\underline{x}) = \int_{S_0 \cup S_R} \psi(\underline{x}) \frac{\partial G(\underline{x}, \underline{x}')}{\partial n(\underline{x}, \underline{x}')} - G(\underline{x}, \underline{x}') \frac{\partial \psi(\underline{x})}{\partial n} dS(\underline{x})$$

If ψ satisfies Sommerfield Radiation Condition (SRC) then

$$\int_{S_0 \cup S_R} \psi(\underline{x}) \frac{\partial G(\underline{x}, \underline{x}')}{\partial n(\underline{x}, \underline{x}')} - G(\underline{x}, \underline{x}') \frac{\partial \psi(\underline{x})}{\partial n} dS(\underline{x}) \rightarrow 0$$

as $R \rightarrow \infty$. Notice also, for $G(\underline{x}, \underline{x}') = -\frac{e^{ik|\underline{x} - \underline{x}'|}}{4\pi|\underline{x} - \underline{x}'|}$ also satisfies

$$\frac{\partial G}{\partial r} = ikG + o(r^{-1})$$

so $\psi = O(r^{-1})$.

$$\psi(\underline{x}') = \int_{S_0} \psi(\underline{x}) \frac{\partial G(\underline{x}, \underline{x}')}{\partial n} - G(\underline{x}, \underline{x}') \frac{\partial \psi(\underline{x})}{\partial n} dS(\underline{x})$$

Since $G(\underline{x}, \underline{x}') = G(\underline{x}', \underline{x})$ so $\psi = \hat{p}$ for scattered field. And swapping $\underline{x}, \underline{x}'$ we have

$$\hat{p}(\underline{x}) = \int_{S_0} G(\underline{x}, \underline{x}_s) \frac{\partial \hat{p}(\underline{x}_s)}{\partial n_s} - \hat{p}(\underline{x}_s) \frac{\partial G(\underline{x}, \underline{x}')}{\partial n_s} dS(\underline{x}_s)$$

$\hat{p} = i\omega\rho_0\hat{\phi}$ implies

$$\frac{\partial \hat{p}}{\partial n_s} := \nabla \hat{p} \cdot \underline{n}_s = i\omega\rho_0 \nabla \hat{\phi} \cdot \underline{n}_s$$

So

$$\frac{\partial \hat{p}}{\partial n_s} = i\omega\rho_0 \hat{u}_n$$

where $\hat{u}_n = \hat{u} \cdot \underline{n}_s$.

$$\hat{p}(\underline{x}) = \int_{S_0} G(\underline{x}, \underline{x}_s) i\omega\rho_0 \hat{u}_n(\underline{x}_s) - \hat{p}(\underline{x}_s) \frac{\partial G(\underline{x}, \underline{x}_s)}{\partial n_s(\underline{x}_s)} dS(\underline{x}_s)$$